

Internship assignment: reproduce a published ML result

Time: 5 working days.

Use AI freely. Claude Code, Copilot, whatever you like.

The paper

Lee, Kim, Chu, Narayan, Le, De, Shirodkar, Lee, Son, Kang, Choi, Jeong, Kim, Yeom, Kim and Kim. "Machine learning-guided optimization of atomic layer deposition process." *Communications Materials* (Nature portfolio), 2026.

- DOI: <https://doi.org/10.1038/s43246-026-01206-w>
- Open access. The supplementary information contains the full dataset, printed as numbers rather than plotted. That is unusual and it is why this paper was chosen.

What the paper does

A neural network predicts three properties of a hafnium oxide (HfOx) film from four process parameters. The film is grown by atomic layer deposition using TDMA-Hf and water.

- **Inputs (4):** deposition temperature (100 to 250 C in 25 C steps), precursor temperature (55 to 75 C in 5 C steps), and x and y position on a 4 inch wafer (9 positions).
- **Outputs (3):** thickness, refractive index, and wet etch rate.
- **Model:** a small dense network, 4 inputs, hidden layers 128-64-32-16-8, ELU activation, 3 outputs.
- **Data:** 315 measured points, split 215 train and 100 test.
- **Headline claim:** above 90 percent accuracy within +/-1.0 nm on thickness, +/-0.04 on refractive index, and +/-0.9 A/min on wet etch rate. They also report comparable performance training on only 100 points.

You do not need any background in semiconductors or chemistry. Everything you need about the process is in the paper. If a term is unfamiliar, look it up and say so in your report.

Assignment

1. Prepare and check the data

Extract the dataset from the supplementary information into a machine-readable file and commit it in the repo. Use any sensible method: conversion tools, OCR, AI-assisted extraction, or manual entry. Do not spend more than half a day on this step.

Run enough structural checks and source spot-checks to establish that the dataset is fit for the analysis. Briefly record what you checked and any uncertainty that remains. We care about whether you detect plausible-looking data errors, not about manual transcription as an end in itself.

2. Reproduce the headline result

Rebuild the model as described and reproduce the accuracy claim. Report what you get.

If your numbers differ from the paper, that is a finding, not a failure. Say by how much and give your best explanation. Do not tune until it matches. Separate choices stated in the paper from choices you had to infer, and report variability across repeated runs or splits where it is relevant.

3. Interrogate the metric

The paper reports “accuracy above 90 percent within ± 1.0 nm”. Work out precisely what that sentence means, then answer:

- What would that number be under a different but equally reasonable definition?
- How much of the reported performance comes from the model, and how much from the tolerance being wide relative to the spread of the data? Design something that separates the two and run it.
- Compare the neural network with at least one simple baseline. Choose the baseline before looking at its result and explain what question it answers.

4. Investigate the errors

Find where the model performs well and where it fails. Test at least one explanation for the pattern you observe. Prefer a small analysis that changes your conclusion over a large collection of plots.

5. Write it up

A short report. Two pages maximum, excluding figures. Aim shorter.

Deliverables

1. **A git repository.** Code, extracted data, and a README that lets us reproduce your results from a clean checkout.
 2. **The report** (max 2 pages). Structure it however you like, but it must contain:
 - what you found, up front
 - **every assumption you made**, listed explicitly, including ones you think are obvious
 - **what you got wrong or could not finish**, and why
 3. **A daily log.** One/two lines per day. AI errors are interesting.
 4. **A 30 minute walkthrough** at the end, live, with no AI assistance. We will pick parts of your code and ask why they are the way they are, then change a requirement and ask you to modify it on the spot.
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What we are looking for

We are not testing whether you can produce a working model. AI will get you most of the way there in a day. We are testing the things it cannot do for you.

1. **Verification against something real.** Did you test important assumptions and results against the source or a meaningful baseline, or accept output that merely looked plausible? Data extraction is one example, not the main task.
2. **Assumptions.**
3. **What failed.**
4. **Knowing what your code does.** In the walkthrough you should be able to explain any line you submitted, and change it.
5. **Scoping.** Five days is not enough to do everything above thoroughly. Prioritize aggressively.

Not evaluated: volume of code, speed, extra features, polish on the parts that were already easy.

Practical notes

- Any language and any libraries.
- Ask questions when things are not clear.
- If something in this brief is ambiguous, resolve it yourself. Write down how you resolved it and move on.